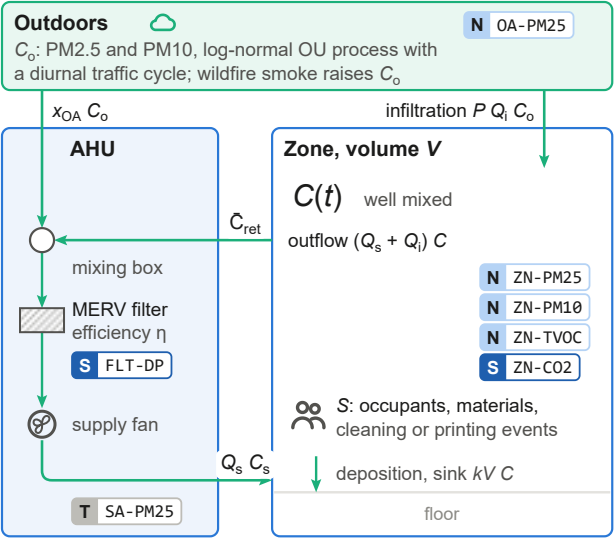


Mass balance for PM2.5, PM2.5–10 and TVOC

$$V\,dC/dt = Q_s C_s + P\,Q_i C_o + S - (Q_s + Q_i + kV)\,C$$

$$C_s = (1 - \eta)(x_{OA} C_o + (1 - x_{OA})\,\bar{C}_{ret})$$

integrated exactly over each 5-min step

Tiers: S standard · A advanced · N next-gen · T truth only

Other zone channels

Illuminance

$E = E_d\,P_{light}/P_{light,d} + K_v\,U\,\Phi_{sol}/A$

lumen method plus daylight flux

A ZN-LUX · A LIGHT-LVL

Sound level

terminal noise $L_d + 50\,\log_{10}(\dot{V}/\dot{V}_{max})$, speech of occupants, background

N ZN-DBA

Thermal comfort

mean radiant and operative temperature, PMV and PPD

N ZN-MRT · N ZN-OPT · N PMV

Occupant feedback

votes round(PMV + $\sigma_i\varepsilon$) at a Poisson rate; hot or cold complaints ∝ N PPD

N TS-VOTE · N HOT-CALLS

Occupancy

stochastic occupant count (truth) seen by a counter and a PIR sensor

N OCC-COUNT · S ZN-OCC